

Gromov's Euclidean Endpoint C^0 Rigidity for the Positive Mass Theorem

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Outline

The problem, background, and history

The Mazurowski–Yao theorem

Endpoint theorem and the new ingredients

References

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The classical rigidity mechanism

The positive mass theorem

Under the standard asymptotically flat hypotheses,

$$\text{Scal}_g \geq 0 \implies m_{\text{ADM}}(g) \geq 0, \quad m_{\text{ADM}}(g) = 0 \implies (\mathbb{R}^3, g) \cong (\mathbb{R}^3, g_{\text{Euc}}).$$

The question at infinity

Can one replace exact Euclidean behavior outside a compact set by a weaker condition visible directly from the metric, while still forcing Euclidean rigidity?

More specifically: can zero-mass rigidity be recognized without derivative decay and without first defining an ADM integral?

Zeroth order: exclude a visibly non-Euclidean cone

A C^0 trace of a second-order condition

Although scalar curvature is a second-order quantity, lower scalar-curvature bounds can persist in C^0 limits. Suppose that a blow-down converges in $C^0_{\text{loc}}(\mathbb{R}^3 \setminus \{0\})$ to a cone smooth away from its vertex. Gromov's C^0 -closure theorem, applied on fixed annuli, gives $\text{Scal} \geq 0$ for the limit cone.

Consider the rotationally symmetric cone

$$g_\alpha = dr^2 + \alpha^2 r^2 g_{S^2}.$$

Its scalar curvature is

$$\text{Scal}_{g_\alpha} = \frac{2(1 - \alpha^2)}{\alpha^2 r^2}.$$

Consequence

If $\alpha > 1$, then $\text{Scal}_{g_\alpha} < 0$. Hence the wider rotational cone g_α cannot occur as a C^0 blow-down limit of a metric with $\text{Scal}_g \geq 0$.

First order: the Schwarzschild mass scale

Once the zeroth-order blow-down is Euclidean, the Schwarzschild family identifies the next asymptotic scale:

$$g_{\text{Sch},m} = \left(1 + \frac{m}{2r}\right)^4 g_{\text{Euc}} = \left(1 + \frac{2m}{r} + O(r^{-2})\right) g_{\text{Euc}}.$$

The endpoint scale

In the Schwarzschild model, the mass parameter first appears at order r^{-1} . Therefore,

$$|g - g_{\text{Euc}}| = o(r^{-1})$$

removes the full Schwarzschild-scale correction, rather than merely the zeroth-order blow-down error.

Why the Euclidean endpoint is harder

A C^0 zero-mass condition

Classically, zero ADM mass forces Euclidean rigidity. The Schwarzschild expansion shows that

$$|g - g_{\text{Euc}}| = o(r^{-1})$$

is the natural C^0 analogue: it removes the entire mass-scale correction.

The new difficulty

No derivative decay is assumed, so the ADM mass is not available a priori. One must prove global Euclidean rigidity directly from this C^0 information at infinity.

Why $o(r^{-1})$ is sharp

A borderline counterexample

Choose a non-round metric $\bar{g}$ on S^3 , round near $p \in S^3$. If G_p is the Green function of $L_{\bar{g}}$, then

$$g_c := G_p^4 \bar{g} \quad \text{on } S^3 \setminus \{p\} \cong \mathbb{R}^3$$

can be chosen smooth, complete, scalar-flat, and non-flat, with

$$g_c = \left(1 + \frac{c}{r} + O(r^{-2})\right) g_{\text{Euc}}, \quad c > 0.$$

Sharpness

Thus

$$\text{Scal}_g \geq 0, \quad |g - g_{\text{Euc}}| = O(r^{-1}) \quad \implies \quad g = g_{\text{Euc}}$$

is false. Even arbitrarily small nonzero r^{-1} -terms allow non-flat metrics; hence $o(r^{-1})$ is the sharp cutoff.

The later literature: a route to the endpoint

Agostiniani–Mazzeri–Oronzio (2024)

They introduce a Green-function monotonicity formula. Their quantity F satisfies

$$\mathrm{Scal}_g \geq 0 \implies F \geq 0, \quad F \text{ is non-decreasing}, \quad F \equiv 0 \implies g = g_{\mathrm{Euc}}.$$

Mazurowski–Yao (2026)

They prove Euclidean rigidity under the stronger decay

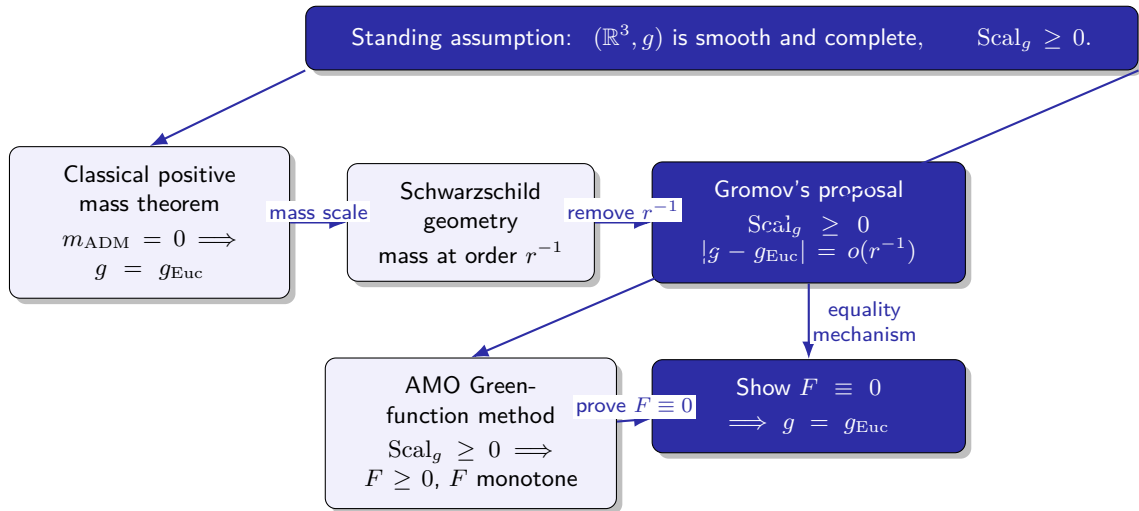
$$|g - g_{\mathrm{Euc}}| = O(r^{-1-\tau}), \quad \tau > 0.$$

You–Zhang (2026)

We remove the extra power of decay and prove the sharp endpoint

$$|g - g_{\mathrm{Euc}}| = o(r^{-1}).$$

Conceptual map: from mass to endpoint rigidity



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The result before the endpoint

Mazurowski–Yao theorem

Let g be a smooth complete metric on $\mathbb{R}^3$ with $\text{Scal}_g \geq 0$. If

$$|g(x) - g_{\text{Euc}}(x)| = O(|x|^{-1-\tau}) \quad \text{for some } \tau > 0,$$

then g is flat.

This proves Gromov's conjecture under a slightly faster decay assumption.

The argument does not first construct an ADM mass. Instead, it converts AMO monotonicity into the vanishing of a stable annular quantity.

The rigidity mechanism: first F , then D

For regular values of $1/t$, define

$$F(t) = 4\pi t - t^2 \int_{\{u=1/t\}} H_g |\nabla_g u| \, dA_g + t^3 \int_{\{u=1/t\}} |\nabla_g u|^2 \, dA_g.$$

Agostiniani–Mazzeri–Oronzio monotonicity

If $\text{Scal}_g \geq 0$, then

$$F(t) \geq 0, \quad F \text{ is non-decreasing.}$$

Moreover,

$$F \equiv 0 \quad \implies \quad (\mathbb{R}^3, g) \text{ is Euclidean.}$$

The difficulty is that F depends on the geometry of Green-function level sets, in particular on their mean curvature. Its continuity is therefore not transparent under merely C^0 convergence of the metric.

Averaging F : the reduction to $aD(a) \rightarrow 0$

Fix $\psi \in C_c^\infty((0, 1))$, $\psi \geq 0$, and $\int_0^1 \psi = 1$. Define

$$E(a, s) = \int_{a+s}^{2a+2s} \frac{F(t)}{t^3} dt, \quad D(a) = \int_0^a \psi(s/a) E(a, s) ds.$$

Inherited monotonicity

$$D(a) \geq 0, \quad a \mapsto aD(a) \text{ is non-decreasing.}$$

The single quantitative target

$$aD(a) \rightarrow 0 \implies D \equiv 0 \implies F \equiv 0 \implies g = g_{\text{Euc}}.$$

The remainder of the proof is devoted to proving $aD(a) \rightarrow 0$.

Put D on a fixed annulus

After rescaling, set

$$u_a(y) = a u(ay), \quad g_a(y) = g(ay), \quad \rho(y) = |y|^{-1}.$$

For large a ,

$$D(a) = \mathcal{D}(g_a, u_a),$$

where

$$\mathcal{D}(\gamma, f) = c_\psi + \int_{\Omega} \varphi(f) |\nabla_\gamma f|^3 dv_\gamma$$

on a fixed annulus $\Omega \Subset \mathbb{R}^3 \setminus \{0\}$.

In the Euclidean model,

$$\mathcal{D}(\delta, \rho) = 0.$$

Thus $aD(a) \rightarrow 0$ becomes a first-order asymptotic question for $\mathcal{D}$ near (δ, ρ) .

First-order reduction: reduce to the Green function

Write

$$k_a := g_a - \delta, \quad v_a := u_a - \rho.$$

Under the endpoint decay assumption,

$$ak_a \longrightarrow 0 \quad \text{in } C^0(\Omega).$$

Thus, in the first-order expansion of $\mathcal{D}(g_a, u_a)$ around the Euclidean model (δ, ρ) , the remaining nontrivial input comes from the rescaled Green-function error v_a .

Remaining analytic task

Obtain a sharp first-order expansion of

$$av_a = a(u_a - \rho)$$

on each fixed annulus $\Omega \in \mathbb{R}^3 \setminus \{0\}$.

The next step is therefore a PDE question: what is the first-order asymptotic term, and how is its coefficient determined?

The Green equation and its source field

Let u be the positive Green function, normalized to have flux 4π . Set

$$A = \sqrt{\det(g_{kl})} g^{-1} = I - B.$$

Then, away from the pole,

$$\operatorname{div}(A\nabla u) = 0 \quad \implies \quad \Delta u = \operatorname{div}(B\nabla u).$$

Choose a smooth vector field X which agrees with $B\nabla u$ outside a compact set. Then

$$\Delta u = \operatorname{div} X \quad \text{near infinity.}$$

Decomposition at infinity

Let w be the Newtonian potential of $\operatorname{div} X$, so that $\Delta w = \operatorname{div} X$ in distribution, and set $h := u - w$. Then h is harmonic outside a compact set and decays at infinity. The asymptotic expansion of u is obtained by combining the large-scale expansion of w with the exterior harmonic expansion of h .

Expansion of the Newtonian potential

Recall that w is defined by

$$w(x) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \left\langle \frac{x-z}{|x-z|^3}, X(z) \right\rangle dz, \quad \Delta w = \operatorname{div} X.$$

If $B = O(r^{-1-\tau})$, then the Green-function and Caccioppoli estimates imply

$$\int_{B_{8R} \setminus B_{R/8}} |X| dx \leq CR^{-\tau} \Rightarrow X \in L^1(\mathbb{R}^3).$$

Hence the following is well defined

$$P_\infty := \frac{1}{4\pi} \int_{\mathbb{R}^3} X(x) dx.$$

Expansion of w

On every fixed annulus $\Omega \Subset \mathbb{R}^3 \setminus \{0\}$,

$$a^2 w(ay) \longrightarrow \frac{\langle P_\infty, y \rangle}{|y|^3} \quad \text{in } W^{1,p}(\Omega), \quad p > 3.$$

The harmonic remainder completes the expansion

The remainder $h = u - w$ is harmonic outside a compact set. Since u has normalized flux 4π , its exterior harmonic expansion takes the form

$$h(x) = \frac{1}{|x|} + \frac{\langle b, x \rangle}{|x|^3} + o(|x|^{-2})$$

for some $b \in \mathbb{R}^3$.

Combining the expansions of w and h , and writing

$$q_\infty := b + P_\infty,$$

one obtains

$$a(u_a - \rho) = a^2(u(ay) - |ay|^{-1}) \longrightarrow \frac{\langle q_\infty, y \rangle}{|y|^3}$$

in $W^{1,p}(\Omega)$, for every $p > 3$.

Outcome

The leading behavior of the Green function is $|x|^{-1}$. The formula above identifies the next term, with a fixed vector q_∞ . After rescaling, this is exactly the part seen by the linearization of $\mathcal{D}$.

Insert the Green expansion into $\mathcal{D}$

The first variation of $\mathcal{D}$ at (δ, ρ) is

$$\begin{aligned} \mathcal{L}(k, v) = & \underbrace{\int_{\Omega} \varphi'(\rho) |\nabla \rho|^3 v \, dy + 3 \int_{\Omega} \varphi(\rho) |\nabla \rho| \langle \nabla \rho, \nabla v \rangle \, dy}_{\mathcal{L}(0, v)} \\ & + \underbrace{\int_{\Omega} \varphi(\rho) \left[\frac{1}{2} (\operatorname{tr} k) |\nabla \rho|^3 - \frac{3}{2} |\nabla \rho| k(\nabla \rho, \nabla \rho) \right] \, dy}_{\mathcal{L}(k, 0)}. \end{aligned}$$

Exact cancellation of first-order translations

For every $q \in \mathbb{R}^3$,

$$\mathcal{L} \left(0, \frac{\langle q, y \rangle}{|y|^3} \right) = 0.$$

Therefore,

$$aD(a) = \mathcal{L}(ak_a, av_a) + o(1) \longrightarrow \mathcal{L}(0, d_{q_\infty}) = 0.$$

Monotonicity now yields $D \equiv 0$, hence $F \equiv 0$, hence flatness.

Where the Mazurowski–Yao method stops

Endpoint obstruction

At $B = o(r^{-1})$,

$$m_j := \int_{2^j \leq |x| \leq 2^{j+1}} |X| \, dx \longrightarrow 0,$$

but this does not force $X \in L^1(\mathbb{R}^3)$.

Thus

$$P_\infty := \frac{1}{4\pi} \int_{\mathbb{R}^3} X \, dx$$

may not exist, so the next correction to $|x|^{-1}$ need not be described by one fixed vector q_∞ .

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Main theorem

Endpoint theorem

Let g be a smooth complete metric on $\mathbb{R}^3$ with $\text{Scal}_g \geq 0$. If

$$|g(x) - g_{\text{Euc}}(x)| = o(|x|^{-1}) \quad \text{as } |x| \rightarrow \infty,$$

then $(\mathbb{R}^3, g)$ is isometric to Euclidean space.

The AMO monotonicity formula and the averaged D -functional remain the rigidity mechanism.
The new issue is to prove

$$aD(a) \longrightarrow 0$$

without the fixed P_∞ available under faster decay.

What remains true at the endpoint

Define the annular modulus

$$\eta(R) = \sup_{R \leq |x| \leq 2R} R|B(x)|.$$

The endpoint decay gives $\eta(R) \rightarrow 0$. Caccioppoli and the Green-function bounds imply

$$\int_{\{R \leq |x| \leq 2R\}} |X| \, dx \leq C\eta(R).$$

Thus the source field X satisfies

$$\int_{\mathbb{R}^3} \frac{|X(x)|}{1 + |x|^2} \, dx < \infty.$$

Endpoint replacement of integrability

Weighted control remains, and the contribution from each large annulus tends to zero.

The Newtonian potential still exists

Set

$$K(z) = \frac{z}{|z|^3}, \quad w(x) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \langle K(x-y), X(y) \rangle \, dy.$$

Endpoint potential

The weighted estimate implies

$$w \in L^1_{\text{loc}}(\mathbb{R}^3), \quad \Delta w = \text{div } X$$

in the distributional sense.

Consequently,

$$h := u - w$$

is harmonic outside a compact set. Thus the usual potential+harmonic splitting of u still survive.

First replacement: from fixed P_∞ to moving P_R^σ

Fix $\sigma > 0$ and a dyadic radius $R = 2^N$. Define

$$P_R^\sigma = \frac{1}{4\pi} \int_{\{|x| \leq \sigma R\}} X(x) \, dx.$$

Growth and compatibility

Along dyadic scales,

$$|P_R^\sigma| = o(\log R).$$

Moreover, for fixed $0 < \sigma_1 < \sigma_2 < \infty$,

$$P_R^{\sigma_2} - P_R^{\sigma_1} \longrightarrow 0.$$

The coefficient may now vary with the observation scale. Although it need not converge, its variation between comparable large scales tends to zero, which is sufficient for the later nonlinear estimate.

Moving- P_R^σ expansion for the potential

Let

$$\Omega_{\lambda,\Lambda} = B_\Lambda \setminus B_\lambda, \quad 0 < \lambda < \Lambda < \infty,$$

and fix $0 < \sigma < \lambda/4$.

Endpoint replacement for the fixed- P_∞ expansion

For every $1 \leq q < 3/2$,

$$\left\| R^2 w(R \cdot) - \frac{\langle P_R^\sigma, y \rangle}{|y|^3} \right\|_{L^q(\Omega_{\lambda,\Lambda})} \longrightarrow 0$$

as $R \rightarrow \infty$ through dyadic scales.

The harmonic remainder

The remainder $h = u - w$ is harmonic outside a compact set and tends to zero at infinity. Hence

$$h(x) = \frac{c}{|x|} + \frac{\langle b, x \rangle}{|x|^3} + O(|x|^{-3}).$$

The 4π -normalization of the Green function, together with the rescaled $W^{1,p}$ convergence, yields

$$c = 1.$$

Thus the harmonic part contributes the fixed term $\frac{\langle b, x \rangle}{|x|^3}$:

$$h(x) - \frac{1}{|x|} = \frac{\langle b, x \rangle}{|x|^3} + O(|x|^{-3}).$$

The endpoint Green expansion: the L^q step

Fix $0 < \sigma < \lambda/8$ and let $R = 2^N$. Define

$$U_R(y) = Ru(Ry), \quad \rho(y) = |y|^{-1}, \quad Q_R^\sigma = b + P_R^\sigma, \quad d_q(y) = \frac{\langle q, y \rangle}{|y|^3}.$$

For $1 \leq q < 3/2$, write

$$\begin{aligned} E_R &:= R(U_R - \rho) - d_{Q_R^\sigma} \\ &= (R^2 w(R \cdot) - d_{P_R^\sigma}) + (R^2 h(R \cdot) - R\rho - d_b). \end{aligned}$$

The first term tends to zero in $L^q(\Omega_{\lambda, \Lambda})$, while the second is $O(R^{-1})$ there. Hence

$$\|E_R\|_{L^q(\Omega_{\lambda, \Lambda})} \longrightarrow 0.$$

Moreover,

$$|Q_R^\sigma| = o(\log R).$$

Elliptic upgrade to $W^{1,p}$

Set

$$A_R(y) = A(Ry), \quad B_R(y) = B(Ry).$$

Using $\operatorname{div}(A_R \nabla U_R) = 0$ and $\Delta \rho = \Delta d_{Q_R^\sigma} = 0$ away from the origin, we obtain

$$\operatorname{div}(A_R \nabla E_R) = \operatorname{div} F_R, \quad F_R = R B_R \nabla \rho + B_R \nabla d_{Q_R^\sigma}.$$

On every slightly larger fixed annulus, there holds $\|B_R\|_{L^\infty} = o(R^{-1})$ and $|\nabla d_{Q_R^\sigma}| \lesssim |Q_R^\sigma|$, and therefore

$$\|F_R\|_{L^\infty} \longrightarrow 0.$$

The analytic output

The sequence estimate upgrades the L^q convergence to

$$\|E_R\|_{W^{1,p}(\Omega_{\lambda,\Lambda})} \longrightarrow 0 \quad \text{for every finite } p > 3.$$

This is the topology needed to expand the D -functional.

Return to the D -functional

Recall that the AMO reduction gives

$$D(a) \geq 0, \quad aD(a) \text{ is non-decreasing.}$$

The preceding $W^{1,p}$ expansion of the rescaled Green function is precisely the input needed to estimate $D(a)$ near the Euclidean pair (δ, ρ) .

The sole remaining target

It is enough to prove

$$aD(a) \longrightarrow 0$$

along dyadic scales. Then

$$D \equiv 0 \implies F \equiv 0 \implies g = g_{\text{Euc}}.$$

Linearization of the D -functional

Set $k_a = g_a - \delta$ and $v_a = u_a - \rho$. The endpoint expansion gives

$$av_a = d_{q_a} + \varepsilon_a, \quad |q_a| = o(\log a), \quad a\|k_a\|_{C^0(\Omega)} + \|\varepsilon_a\|_{W^{1,p}(\Omega)} \longrightarrow 0.$$

Expanding at (δ, ρ) ,

$$\begin{aligned} \mathcal{D}(\delta + k_a, \rho + v_a) &= L_D(k_a, v_a) + \mathcal{R}_a, \\ aL_D(k_a, v_a) &= aL_D(k_a, 0) + L_D(0, d_{q_a}) + L_D(0, \varepsilon_a). \end{aligned}$$

Exact cancellation at first order

For every $q \in \mathbb{R}^3$,

$$L_D(0, d_q) = 0.$$

Hence

$$aL_D(k_a, v_a) \longrightarrow 0.$$

It remains only to show

$$a\mathcal{R}_a \longrightarrow 0.$$

Why a quadratic remainder estimate is needed

Qualitative differentiability gives only

$$\mathcal{R}_a = o(\|k_a\|_{C^0} + \|v_a\|_{W^{1,p}}).$$

But

$$a\|k_a\|_{C^0} \rightarrow 0, \quad a\|v_a\|_{W^{1,p}} \leq C|q_a| + o(1),$$

and $|q_a|$ need not remain bounded. Thus this does not imply $a\mathcal{R}_a \rightarrow 0$.

Quantitative second-order control

Instead, we need

$$|\mathcal{R}(k, v)| \leq C(\|k\|_{C^0} + \|v\|_{W^{1,p}})^2.$$

Since

$$\|k_a\|_{C^0} + \|v_a\|_{W^{1,p}} = o\left(\frac{\log a}{a}\right),$$

we obtain

$$a|\mathcal{R}_a| = o\left(\frac{(\log a)^2}{a}\right) \rightarrow 0.$$

Monotonicity closes the proof

Recall that

$$D(a) \geq 0, \quad aD(a) \text{ is non-decreasing.}$$

For every fixed $a > 0$ and every sufficiently large dyadic R ,

$$0 \leq aD(a) \leq RD(R).$$

Since $RD(R) \rightarrow 0$ along dyadic scales, we obtain

$$D(a) = 0 \quad \text{for every } a > 0.$$

Hence

$$F \equiv 0.$$

Rigidity

The equality case of the AMO monotonicity formula gives

$$(\mathbb{R}^3, g) \cong (\mathbb{R}^3, g_{\text{Euc}}).$$

What changes at the endpoint proof?

Mazurowski–Yao

- $g - \delta = O(r^{-1-\tau})$ implies $X = B\nabla u \in L^1$.
- The coefficient

$$P_\infty = \frac{1}{4\pi} \int_{\mathbb{R}^3} X$$

is fixed.

- The rescaled Green expansion is governed by the fixed vector P_∞ .
- Qualitative Fréchet differentiability of $\mathcal{D}$ is sufficient.

Endpoint proof

- $g - \delta = o(r^{-1})$ gives only dyadic masses $m_j \rightarrow 0$.
- The full integral of X may fail to converge, so one uses P_R^σ .
- The coefficient

$$Q_R^\sigma = b + P_R^\sigma$$

may vary with the scale.

- Since $L_D(0, d_q) = 0$ for every q , the linear term still vanishes; a quadratic remainder estimate controls the scale dependence.

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Thank you!